

| Framework ID | Screening question                                                                                    | Purpose                                                                                                               | What to check                                                                                                                                                         | Evidence from project                                                 | Risk if answer is weak                                                                 | Output for decision-maker                         |
|--------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------|
| Q1           | What function must the window system preserve over the study period?                                  | To define the performance that must remain equivalent when comparing repair, replacement or life-extension strategies | Thermal performance, acoustic performance, safety, weather protection, operation, durability and user comfort                                                         | Functional unit from lifecycle scenario comparison                    | Strategies may be compared unfairly if they do not deliver the same function           | Clear functional unit before selecting a strategy |
| Q2           | Which components are likely to fail before the whole window reaches end of life?                      | To distinguish long-life components from short-life wear components                                                   | Hardware, hinges, handles, fittings, seals, gaskets, glazing unit, coatings and drainage details                                                                      | Component map and both expert interviews                              | Full replacement may be chosen even when only a smaller component has failed           | Component-level failure map                       |
| Q3           | Can short-life components be replaced independently without replacing the complete window?            | To test whether component replacement is technically possible                                                         | Independent access, removable parts, replaceable glazing unit, available fittings, documented repair sequence and compatibility                                       | VELFAC case documentation, supplier interview and architect interview | Product-life extension remains theoretical if components cannot be replaced separately | Technical repairability judgement                 |
| Q4           | Is repair practically enabled by access, installation method and maintenance infrastructure?          | To test whether technical repairability can happen in real project conditions                                         | Physical access, wall and window interface, installation detailing, service clearance, skilled labour, maintenance guidance and local service availability            | Interview memo and component map                                      | Repair may fail in practice even when components are technically repairable            | Implementation readiness judgement                |
| Q5           | Which actors must support the strategy over time?                                                     | To identify who must keep the strategy functional during the window life cycle                                        | Manufacturer, supplier, installer, architect, client, maintenance provider, building manager, recycler and possible reuse actor                                       | Interviews and risk pathways                                          | Responsibility gaps may make repair or reuse impossible later                          | Actor responsibility map                          |
| Q6           | What trade-offs are introduced by the circular strategy?                                              | To avoid assuming that repairability, modularity or high performance is automatically beneficial                      | Added material, added connectors, technical complexity, cost, repeated service visits, transport, labour and possible future upgrade limits                           | Lifecycle scenario comparison and transparent assumptions             | Expected benefits may be reduced by production, maintenance or transport burdens       | Trade-off statement                               |
| Q7           | Could the strategy trigger rebound, displacement or unintended replacement behaviour?                 | To check whether the intervention changes behaviour or shifts impacts elsewhere                                       | More frequent upgrading, client preference for newer products, burden shifting to transport or end of life, delayed inefficient performance, or reuse barriers        | Classified risk pathways                                              | Circular design may create hidden or displaced environmental consequences              | Rebound and unintended-consequence screen         |
| Q8           | Is the expected lifecycle benefit still plausible after assumptions and uncertainties are considered? | To make the final decision cautious and evidence-based                                                                | Frame life, glazing life, hardware replacement interval, repair frequency, spare-part availability, transport distance, end-of-life route and operational performance | Transparent assumptions table and lifecycle scenario comparison       | A strategy may be presented as sustainable without enough evidence                     | Final screening conclusion                        |